National Transportation Safety Board
Aviation Accident Final Report

Location:	Atwater, CA	**Accident Number:**	SEA08CA030
Date & Time:	11/17/2007, 1414 PST	**Registration:**	N3318E
Aircraft:	Cessna 172N	**Aircraft Damage:**	Substantial
Defining Event:		**Injuries:**	1 Minor
Flight Conducted Under:	Part 91: General Aviation - Instructional		

Analysis

This was the student pilot's first supervised solo. After making his second landing and during the landing roll, the airplane began to veer to the left. An attempt to correct back to the right was unsuccessful, which was followed by the airplane continuing to the left and impacting a runway sign. The airplane continued northwest-bound before impacting a drainage ditch approximately 250 feet from the edge of the runway. The aircraft subsequently turned eastbound and crossed the drainage ditch a second time, which resulted in the collapse of the right main landing gear. The airplane came to rest in an upright position with its tail section in the drainage ditch and its right wing tip laying on the ground. The airplane's right wing sustained substantial damage.

Probable Cause and Findings

The National Transportation Safety Board determines the probable cause(s) of this accident to be: The failure of the student pilot to maintain directional control during the landing roll. Factors contributing to the accident included the sign and the ditch.

Findings

Occurrence #1: LOSS OF CONTROL - ON GROUND/WATER
Phase of Operation: LANDING - ROLL

Findings
1. (C) DIRECTIONAL CONTROL - NOT MAINTAINED - PILOT IN COMMAND

Occurrence #2: ON GROUND/WATER COLLISION WITH OBJECT
Phase of Operation: LANDING - ROLL

Findings
2. (F) OBJECT - SIGN

Occurrence #3: ON GROUND/WATER ENCOUNTER WITH TERRAIN/WATER
Phase of Operation: LANDING - ROLL

Findings
3. (F) TERRAIN CONDITION - DITCH

Factual Information

Student Pilot Information

Certificate:	Student	**Age:**	18, Male
Airplane Rating(s):	Single-engine Land	**Seat Occupied:**	
Other Aircraft Rating(s):	None	**Restraint Used:**	Seatbelt, Shoulder harness
Instrument Rating(s):	None	**Second Pilot Present:**	
Instructor Rating(s):		**Toxicology Performed:**	
Medical Certification:	Class 1	**Last FAA Medical Exam:**	11/01/2007
Occupational Pilot:		**Last Flight Review or Equivalent:**	
Flight Time:	23 hours (Total, all aircraft), 23 hours (Total, this make and model), 15 hours (Last 90 days, all aircraft), 11 hours (Last 30 days, all aircraft)		

Aircraft and Owner/Operator Information

Aircraft Make:	Cessna	**Registration:**	N3318E
Model/Series:	172N	**Aircraft Category:**	Airplane
Year of Manufacture:		**Amateur Built:**	No
Airworthiness Certificate:	Normal	**Serial Number:**	17271502
Landing Gear Type:	Tricycle	**Seats:**	
Date/Type of Last Inspection:		**Certified Max Gross Wt.:**	
Time Since Last Inspection:		**Engines:**	1 Reciprocating
Airframe Total Time:		**Engine Manufacturer:**	Lycoming
ELT:		**Engine Model/Series:**	O-320
Registered Owner:	American School of Aviation	**Rated Power:**	
Operator:	American School of Aviation	**Operating Certificate(s) Held:**	None

Meteorological Information and Flight Plan

Conditions at Accident Site:	Visual Conditions	**Condition of Light:**	Day
Observation Facility, Elevation:	MER	**Distance from Accident Site:**	
Observation Time:	1347	**Direction from Accident Site:**	
Lowest Cloud Condition:	Clear	**Visibility**	10 Miles
Lowest Ceiling:	None	**Visibility (RVR):**	
Wind Speed/Gusts:	6 knots /	**Turbulence Type Forecast/Actual:**	/
Wind Direction:	300°	**Turbulence Severity Forecast/Actual:**	/
Altimeter Setting:	30.07 inches Hg	**Temperature/Dew Point:**	21°C / 13°C
Precipitation and Obscuration:	No Obscuration; No Precipitation		
Departure Point:	Atwater, CA (MER)	**Type of Flight Plan Filed:**	None
Destination:	Atwater, CA (MER)	**Type of Clearance:**	None
Departure Time:	PST	**Type of Airspace:**	

Airport Information

Airport:	Castle Airport (MER)	**Runway Surface Type:**	Asphalt
Airport Elevation:		**Runway Surface Condition:**	Dry
Runway Used:	31	**IFR Approach:**	None
Runway Length/Width:	11802 ft / 150 ft	**VFR Approach/Landing:**	Traffic Pattern

Wreckage and Impact Information

Crew Injuries:	1 Minor	**Aircraft Damage:**	Substantial
Passenger Injuries:	N/A	**Aircraft Fire:**	None
Ground Injuries:	N/A	**Aircraft Explosion:**	
Total Injuries:	1 Minor	**Latitude, Longitude:**	37.380556, -120.568056

Administrative Information

Investigator In Charge (IIC):	Thomas M Little	**Report Date:**	01/31/2008
Additional Participating Persons:			
Publish Date:			
Note:	This accident report documents the factual circumstances of this accident as described to the NTSB.		
Investigation Docket:	NTSB accident and incident dockets serve as permanent archival information for the NTSB's investigations. Dockets released prior to June 1, 2009 are publicly available from the NTSB's Record Management Division at pubinq@ntsb.gov, or at 800-877-6799. Dockets released after this date are available at http://dms.ntsb.gov/pubdms/.		

The National Transportation Safety Board (NTSB), established in 1967, is an independent federal agency mandated by Congress through the Independent Safety Board Act of 1974 to investigate transportation accidents, determine the probable causes of the accidents, issue safety recommendations, study transportation safety issues, and evaluate the safety effectiveness of government agencies involved in transportation. The NTSB makes public its actions and decisions through accident reports, safety studies, special investigation reports, safety recommendations, and statistical reviews.

The Independent Safety Board Act, as codified at 49 U.S.C. Section 1154(b), precludes the admission into evidence or use of any part of an NTSB report related to an incident or accident in a civil action for damages resulting from a matter mentioned in the report. A factual report that may be admissible under 49 U.S.C. § 1154(b) is available here.